Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Chiral Plasmonic–Enhanced Optical Diagnostic Platform for Label-Free, Chirality-Resolved Biosensing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio·medical		*Chemicals*	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic Devices & Diagnostic Devices	*Bio-signal Measurement / Diagnostic Devices*	*Fine Chemicals*	*Nano-Application Technologies*
Concept of Proposed Technology	The proposed technology is a chiral plasmonic–enhanced optical diagnostic platform that enables label-free, chirality-resolved biosensing without reliance on conventional circularly polarized light (CPL) or complex polarization-control optics.[1–3] Rather than inferring molecular properties indirectly through refractive index or intensity changes, the platform is designed to directly transduce molecular chirality, stereochemical composition, and higher-order structural asymmetry into optical signals at the sensor-module level. At the core of the platform are engineered chiral plasmonic nanostructures that intrinsically generate and amplify chiroptical responses. These nanostructures form localized superchiral electromagnetic fields within the near-field region, enabling selective interactions with chiral and structurally asymmetric biomolecules.[4–8] As a result, chiroptical contrast is generated directly at the material and nano–bio interface levels, without external polarization modulation. This intrinsic signal-generation mechanism allows optical readout using simplified optical configurations, ensuring compatibility with compact, modular diagnostic systems. The platform is designed for direct integration into in vitro diagnostic (IVD) systems, bioanalytical and life science instrumentation, pharmaceutical quality assessment platforms, and advanced chemical and materials analysis systems.[9–12] Emphasizing scalability, reproducibility, and translational deployment, the proposed technology establishes a new class of optical diagnostic modules that redefine how chiroptical information is generated, read, and interpreted beyond laboratory-scale spectroscopic techniques.			

Rationale for Joint R&D Project	Unlike conventional plasmonic sensors, which primarily rely on changes in analyte concentration or refractive index, chiral plasmonic platforms provide direct access to molecular asymmetry and structural handedness, enabling intrinsically label-free biosensing that can capture subtle structural and conformational alterations associated with disease states, therapeutic response, or pathological progression. The proposing research group has established strong expertise in the synthesis of chiral plasmonic nanostructures,[13–19] quantitative chiroptical characterization,[18,20–23] and biosensing applications,[24,25] supported by a solid record of both fundamental and applied research in chiral nanophotonics.[18,19,26,27] Through prior studies, the group has demonstrated the feasibility of using chiral plasmonic platforms to optically resolve disease-associated biomolecular variations, thereby providing a robust scientific foundation for the proposed research. However, advancing chirality-resolved optical diagnostics from laboratory-scale proof-of-concept demonstrations to module-integrated, reproducible sensing systems at TRL 4–5 requires tightly integrated development across optical hardware engineering, signal processing, and system-level validation. The Korean SME contributes complementary expertise in optical module optimization, system integration, manufacturable hardware design, and embedded signal processing, enabling the translation of complex chiroptical phenomena into stable, user-ready diagnostic outputs. Through the co-development of simplified optical configurations, instrument level preprocessing, and modular hardware architectures, this partnership directly addresses key translational bottlenecks including repeatability, robustness, and operational stability, and establishes a device oriented, translational technology foundation suitable for future diagnostic applications.
Key Words	1 Plasmonic nanomaterials 2. Chirality 3. Biosensors 4. Optical signal processing

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (2-3)	**End of the Project**	Level (4-5)
R&D Objectives	Optical sensing technologies are widely implemented as miniaturized, device integrated sensor modules across clinical diagnostics, life science instrumentation, pharmaceutical analysis, and advanced chemical and materials characterization.[28,29] Recent efforts have moved beyond intensity based or binary detection toward interpreting molecular structural, stereochemical, and conformational variations as continuous analytical states, enabling earlier detection and improved diagnostic interpretability. However, conventional chiroptical techniques rely on circularly polarized light (CPL) and complex polarization optics, fundamentally limiting integration into compact sensor modules and practical diagnostic systems.[12,30–32] The central goal of this project is to develop a salable chiroptical sensing architecture that enables direct and reproducible optical transduction of molecular chirality and structural asymmetry without dependence on complex polarization-control optics. To this end, the project will engineer chiral plasmonic nanostructures that intrinsically generate and amplify chiroptical contrast or chirality-dependent optical anisotropy at the material and nano–bio interface levels. By translating these effects into stable sensor outputs that can be read using simplified optical configurations, this work establishes a fundamentally new approach to module level chiroptical sensing. Building upon this sensing architecture, the project further aims to establish a state resolved sensing and readout framework that interprets molecular structural and stereochemical changes as continuous states or analytical axes rather than binary detection outcomes. This capability will enable sensitive tracking of protein conformational changes and aggregation behavior, stereochemical composition of chiral molecules, and other structure-dependent molecular			

	variations.[23–25,33] Ultimately, the proposed framework is designed for direct integration into in vitro diagnostic systems, bioanalytical and life science instrumentation, pharmaceutical quality control platforms, and advanced chemical and materials analysis systems, emphasizing translational deployment rather than extension of laboratory-scale spectroscopic techniques.
R&D Target Specifications	The ultimate target of this R&D effort is to establish a chiral plasmonic biosensing core technology that combines high analytical sensitivity, structure-resolved selectivity, and operational robustness, enabling translation toward in vitro diagnostic (IVD) and bioanalytical applications. To support this objective, the following technical targets and performance specifications are defined in alignment with the phased R&D plan: **1. Chiral Plasmonic Sensing Materials and Hardware Platform** • **Material Platform Development**: Development of chiral plasmonic sensing materials based on noble metals (e.g., gold, silver) for optical biosensing applications. • **Chiroptical Activity**: Achievement of high chiroptical activity, with an initial target of g-factor ≥ 0.03 during early hardware validation, and systematic optimization toward g-factor ≥ 0.05 in later phases. • **Analytical Sensitivity**: Demonstration of high analytical sensitivity, targeting sub-picomolar (≤ pM) detection limits for disease-relevant biomarkers. • **Optical and Sensing Stability**: Verification of robust optical stability and sensing performance under repeated measurements and extended operation, targeting coefficient of variation (CV) ≤ 10% across ≥ 10 repeated measurements under identical conditions • **Surface Functionalization**: Implementation of biomarker-specific surface functionalization strategies (e.g., small molecules, peptides, proteins, antibodies) to enhance selectivity and suppress non-specific interactions. **2. Chiroptical Signature, Selectivity, and Diagnostic Performance** • **Signature Establishment**: Establishment of distinct and reproducible chiroptical signatures capable of achieving ≥80% selectivity (discrimination reliability) between control and biomarker-positive samples. • **Specificity Validation**: Validation of target specificity and signature robustness across diverse biomarker types, including protein mutations, presence/absence of target biomolecules, post-translational modifications (PTMs), and protein aggregation states. • **Advanced Data Analysis**: Development of optical signal processing and data analysis methodologies to further enhance diagnostic performance, with a target diagnostic accuracy approaching ~ 90%. • **Operational Reliability**: Verification of signal stability, reproducibility, and interpretability across multiple measurement cycles.
R&D Plan	**1. Phase I: Establishment of the Chiroptical Hardware Core and Validation with Tier 1 Targets (0–6 months)** During the first six months, the project will focus on establishing a **high-sensitivity chiral plasmonic biosensing hardware core** suitable for bio-detection. Chiral plasmonic nanostructures will be designed and fabricated to achieve a **g-factor ≥ 0.03**, and **polarization-sensitive optical readout schemes** will be introduced to realize **picomolar-level detection sensitivity** under controlled conditions. In parallel, **biomolecule-friendly surface functionalization strategies** will be developed to ensure stable target binding while suppressing nonspecific background signals. • Design and implementation of the sensing hardware core architecture incorporating chiral plasmonic and chiral optically active elements • CD-based measurements to elucidate the physical origin of chiral signals and to assess the influence of hardware components (polarization, alignment, and optical configuration)

- For SERS-based measurements, establishment of the basic optical configuration (light source, detector, filters, and optics) and measurement protocols for chiral anisotropy readout
- Verification of repeatability and signal stability through repeated measurements using Tier 1 model targets (e.g., simple chiral molecules and peptides)

Deliverable: TIPA joint R&D Full Proposal; proof-of-concept (PoC) data for the hardware core (for publications and patents)
Milestone: Stable detection of chiral anisotropy signals from Tier 1 targets and establishment of repeatability metrics (e.g., stabilization of repeated measurements and readout indicators)

2. Phase II: Optimization of Optical Modules, Signal Acquisition, and Hardware–Software Integration (7–18 months)

During this phase, the project will focus on optimizing the optical hardware modules and integrating signal acquisition with data processing pipelines to enable robust and repeatable chiroptical readout at the sensor-module level. Optical components, including light sources, detectors, filters, and alignment schemes, will be refined to improve signal stability, while instrument-embedded preprocessing and feature extraction routines will be implemented to convert raw optical responses into reliable readout metrics. In parallel, surface functionalization and sample preparation protocols will be optimized to ensure robustness against variations in sample matrices and binding configurations.

- Enhancement of signal stability at the hardware level, including optimization of optical components, detectors, light sources, and alignment stability
- Implementation of an instrument-embedded signal processing pipeline incorporating preprocessing (background correction, noise suppression, normalization) and feature extraction
- Optimization of surface functionalization and sample preparation conditions for chiral plasmonic structures to ensure robustness against variations in sample matrices
- Validation of architectural generality by expanding the Tier 1 scope to include diverse chiral structures and binding modes

Deliverable: Submission of a joint research publication focusing on hardware implementation and readout metrics
Milestone: Completion of an integrated hardware–software readout protocol (modularized measurement and analysis routines) and enhanced repeatability

3. Phase III: Extension to an Alzheimer's Reference Scenario for Structural-State Readout (19–24 months)

In this phase, the optimized chiroptical sensing architecture will be extended to biologically relevant protein systems associated with Alzheimer's disease, serving as a representative reference scenario for structure-sensitive analysis. Protein systems exhibiting distinct structural states—such as modified or aggregated tau, different ApoE isoforms, and multiple Aβ species—will be analyzed to establish correlations between structural-state differences and chiroptical readout signatures. CD-based analysis will be used to rationalize the origin of structure-dependent signals, while SERS-based measurements will quantify corresponding changes in chiral anisotropy, enabling the construction of a reproducible and selective chiroptical signal panel across multiple biomarkers.

- Selection of Alzheimer's disease–related protein systems as representative reference scenarios and execution of structure-state–based analyses
  - Examples include modified and aggregated tau states, comparison of ApoE isoforms, and differentiation of Aβ species
- Systematic correlation of structural-state changes with chiroptical signals through CD-based analysis, and quantification of **chiral anisotropy variations associated with**

structural changes using SERS-based readout

- Establishment of consistent relationships between "structural-state differences → readout metric differences" for at least 2–3 representative cases; construction of a **chiroptical signal panel** with reproducibility and selectivity across **five or more biomarkers**, and organization of datasets to support subsequent translational signal analysis and model development. A **selectivity ≥ 80%** is targeted at this stage.

Deliverable: Publication (or submission) of a joint research paper; domestic patent application covering hardware and readout architectures

Milestone: Acquisition and reproducibility validation of differential readout metrics reflecting structural-state changes in Alzheimer's reference systems, with **picomolar detection sensitivity**, **g-factor ≥ 0.03**, and **selectivity ≥ 80%,** with pathway toward diagnostic accuracy approaching ~90% through advanced signal analysis

4. Phase IV: Expansion of Sample Matrices and Enhancement of Readout Robustness (25–36 months)

This phase will emphasize enhancing the robustness and reliability of chiroptical readout across increasingly complex sample environments. The sensing platform will be evaluated under a range of matrix conditions, from simple buffer solutions to mixed-component samples and limited biofluid environments, to assess stability and performance consistency. To support reliable operation under these conditions, calibration strategies and correction logic addressing optical drift, background variation, and scattering effects will be further refined, and operational parameters will be standardized from a device-use perspective.

- Evaluation of readout stability across increasingly complex sample matrices (buffer solutions → simple mixtures → limited biofluid conditions)
- Advancement of calibration and correction logic for hardware, protocols, and analysis pipelines to compensate for optical drift, background variation, and scattering effects
- Standardization of readout metrics from an operational perspective, including measurement time, number of repetitions, and decision criteria

Deliverable: Joint research publication and domestic patent(s), including signal processing and readout algorithms

Milestone: Completion of a standardized readout pipeline (standard operating protocol) that operates reliably across varying matrix conditions

5. Phase V: Demonstration of a Modular Hardware Platform at TRL 4–5 (31–36 months)

In the final phase, the project will focus on demonstrating a modular chiroptical sensing hardware platform at the TRL 4–5 level, integrating the validated sensing architecture into a prototype-level system. A modular configuration consisting of an optical head and detection/analysis modules will be defined, and system-level performance—including repeatability, reliability, and operational stability—will be evaluated under laboratory or laboratory-relevant environments. In parallel, productization considerations such as manufacturing feasibility, packaging strategies, and potential use scenarios (e.g., benchtop analytical instruments or cartridge-based measurement formats) will be assessed to inform follow-on commercialization and joint development pathways.

- Definition of a prototype-level configuration for a **modular hardware platform** (optical head plus detection/analysis modules)
- Demonstration of system-level performance, including repeatability, reliability, and operational robustness
- Assessment of productization potential from the perspectives of manufacturing, packaging, and use scenarios (e.g., benchtop analytical instruments, cartridge-based measurement formats)

	Deliverable: Prototype-level system configuration proposal (or prototype) and a roadmap for subsequent commercialization and joint development **Milestone:** Achievement of **TRL 4–5**, defined as verification of the core functions of an integrated hardware platform in a laboratory environment

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The Research Institute will lead the scientific foundations of chiroptical sensing, including chiral plasmonic design, nano–bio interface engineering, and biological interpretation of chiroptical signals. Its responsibilities include: • **Development of chiroptical sensing principles and chiral plasmonic nanostructures** Design and fabrication of chiral plasmonic nanostructures with controlled optical activity, targeting robust chiroptical contrast (g-factor ≥ 0.03), and establishment of the underlying sensing mechanisms. • **Nano–bio interface engineering and surface functionalization strategies** Development of biomolecule-compatible surface chemistries that enable stable target binding while minimizing nonspecific background signals, ensuring preservation of chiroptical signal fidelity. • **Selection of analytical targets and structural-state interpretation** Definition of Tier 1 model targets (e.g., simple chiral molecules and peptides) and biologically relevant protein systems, including Alzheimer's disease–related targets, and interpretation of correlations between structural-state changes and chiroptical responses.
Expected R&R of Korean SME Partner	The Korean SME Partner will lead the **hardware implementation and system integration** of the chiroptical sensing architecture. Its responsibilities include: • **Design and fabrication of optical sensor modules and hardware architectures** Implementation of chiroptical sensing architectures in the form of spectrometer-based sensor modules or compact optical sensor devices, including optical layout, mechanical design, and component integration. • **Development of signal acquisition, stabilization, and readout pipelines** Implementation of instrument-embedded signal processing workflows, including background correction, noise suppression, normalization, and feature extraction, to convert raw optical signals into actionable sensor outputs. • **System integration and TRL 4–5 validation** Integration of sensing principles and interfaces provided by the Research Institute into functional sensor-module prototypes, followed by verification of core functionalities at the TRL 4–5 level in laboratory or laboratory-relevant environments. • **Product-oriented evaluation and scalability assessment** Assessment of manufacturability, packaging strategies, and operational scenarios to inform follow-on commercialization and joint development roadmaps.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The primary outcome of this project is the realization of a hardware-validated chiroptical sensing platform at the sensor-module level, reaching TRL 4–5. By the end of the project, the proposed sensing architecture will be implemented as spectrometer-based sensor modules or compact optical sensor devices, demonstrating reproducible and device-compatible performance. The project will deliver integrated sensing architectures, standardized readout metrics, and embedded signal-processing pipelines, enabling structure-sensitive and chirality-resolved molecular

	detection beyond conventional intensity- or concentration-based optical sensors. From an industrial perspective, the developed sensor-module platform is applicable across multiple sectors, including in vitro diagnostics (IVD), optical metrology and spectroscopy, life science research instrumentation, pharmaceutical and biotechnology quality analysis, and advanced materials and chemical industries. The modular architecture allows low-barrier integration into existing optical platforms, supporting product differentiation, system miniaturization, and scalable deployment. Overall, this project will transition chiroptical plasmonic sensing from laboratory demonstrations to an industry-relevant sensor hardware technology, providing a practical foundation for follow-on commercialization and joint technology development.

6. References

1. Hwang, M. & Yeom, B. Fabrication of Chiral Materials in Nano- And Microscale †. *Chemistry of Materials* **33**, 807–817 (2021).

2. Lininger, A. *et al.* Chirality in Light–Matter Interaction. *Advanced Materials* **35**, 2107325 (2022).

3. Lee, S. *et al.* Unraveling the Chirality Transfer from Circularly Polarized Light to Single Plasmonic Nanoparticles. *Angewandte Chemie - International Edition* **63**, (2024).

4. Kakkar, T. *et al.* Superchiral near fields detect virus structure. *Light Sci. Appl.* **9**, 195 (2020).

5. Tabouillot, V. *et al.* Near-Field Probing of Optical Superchirality with Plasmonic Circularly Polarized Luminescence for Enhanced Bio-Detection. *ACS Photonics* **9**, 3617–3624 (2022).

6. Tullius, R. *et al.* Superchiral Plasmonic Phase Sensitivity for Fingerprinting of Protein Interface Structure. *ACS Nano* **11**, 12049–12056 (2017).

7. Hendry, E. *et al.* Ultrasensitive detection and characterization of biomolecules using superchiral fields. *Nat. Nanotechnol.* **5**, 783–787 (2010).

8. Hwang, M. *et al.* Lithography-Free Fabrication of Terahertz Chiral Metamaterials and Their Chirality Enhancement for Enantiomer Sensing. *Adv. Opt. Mater.* **2300045**, 1–9 (2023).

9. Dixon, A. R. *et al.* Recent developments in multiplexing techniques for immunohistochemistry. *Expert Rev. Mol. Diagn.* **15** 1171–1186 (2015).

10. John, N., Mariamma, A.T. Recent developments in the chiroptical properties of chiral plasmonic gold nanostructures *Microchim. Acta.* **188**, 424 (2021).

11. Dixon, T. A., Williams, T. C. & Pretorius, I. S. Sensing the future of bio-informational engineering. *Nat. Commun.* **12**, 388 (2021).

12. Cho, I.-H., Kim, D. H. & Park, S. Electrochemical biosensors: perspective on functional nanomaterials for on-site analysis. *Biomater. Res.* **24**, 6 (2026).

13. Lu, J., Jung, H. J., Kim, J.-Y. & Kotov, N. A. Bright, circularly polarized black-body radiation from twisted nanocarbon filaments. *Science* **386**, 1400–1404 (2024).

14. Kim, J. Y. *et al.* Assembly of Gold Nanoparticles into Chiral Superstructures Driven by Circularly Polarized Light. *J. Am. Chem. Soc.* **141**, 11739–11744 (2019).

15. Park, K. H. *et al.* Broad Chiroptical Activity from Ultraviolet to Short-Wave Infrared by Chirality Transfer from Molecular to Micrometer Scale. *ACS Nano* **15**, 15229–15237 (2021).

16. Kim, J. Y. *et al.* Direct-write 3D printing of plasmonic nanohelicoids by circularly polarized light. *Proc. Natl. Acad. Sci. U. S. A.* **121**, (2024).

17. Guo, J. *et al.* Broadband Circular Polarizers via Coupling in 3D Plasmonic Meta-Atom Arrays. *ACS Photonics* **8**, 1286–1292 (2021).

18. Kumar, P. *et al.* Photonically active bowtie nanoassemblies with chirality continuum. *Nature* **615**, 418–424 (2023).

19. Ohnoutek, L. *et al.* Third-harmonic Mie scattering from semiconductor nanohelices. *Nat. Photonics* **16**, 126–133 (2022).

20. Hwang, M., Jung, H. & Kim, J.-Y. Chirality Quantification for High-Performance Nanophotonic Biosensors. *Small Methods* **n/a**, 2500112 (2025).

21. Cha, M., Ma, J., Kim, J.-Y., Emre, E. S. T. & Kotov, N. A. Graph-theoretical chirality measure and chirality–property relations for chemical structures with multiscale mirror asymmetries. *Chirality* **36**, e23678 (2024).

22. Kuznetsova, V. *et al.* Graph–Property Relationships for Complex Chiral Nanodendrimers. *ACS Nano* **19**, 6095–6106 (2025).

23. Cha, M. *et al.* Unifying structural descriptors for biological and bioinspired nanoscale complexes. *Nat. Comput. Sci.* **2**, 243–252 (2022).

24. Xu, L. *et al.* Enantiomer-dependent immunological response to chiral nanoparticles. *Nature* **601**, 366–373 (2022).

25. Kang, Y.-T. *et al.* Chiroptical detection and mutation analysis of cancer-associated extracellular vesicles using microfluidics with oriented chiral nanoparticles. *Matter* **7**, 4373–4389 (2024).

26. Kim, J. Y. & Kotov, N. A. Origin of chiroptical activity in nanorod assemblies. *Science* **365** 1378–1379 (2019).

27. Choi, W. J. *et al.* Chiral phonons in microcrystals and nanofibrils of biomolecules. *Nat. Photonics* **16**, 366–373 (2022).

28. Hu, J., Liu, F., Chen, Y., Shangguan, G. & Ju, H. Mass Spectrometric Biosensing: A Powerful Approach for Multiplexed Analysis of Clinical Biomolecules. *ACS Sensors* **6** 3517–3535 (2021).

29. Yanik, A. A. *et al.* An optofluidic nanoplasmonic biosensor for direct detection of live viruses from biological media. *Nano Lett.* **10**, 4962–4969 (2010).

30. Gao, W. *et al.* Recent advances in biosensors: structure, principles, classification, and application in bio-manufacturing. *Methods* **242**, 123–142 (2025).

31. Pisani, M. & Carbonell, P. Challenges and Opportunities in Smart Biosensing for Biomanufacturing. *ACS Synth. Biol.* **14**, 3267–3274 (2025).

32. Herkert, E. K. & Garcia-Parajo, M. F. Harnessing the Power of Plasmonics for in Vitro and in Vivo Biosensing. *ACS Photonics* **12**, 1259-1275 (2025).

33. Hao, C., Xu, L., Kuang, H. & Xu, C. Artificial Chiral Probes and Bioapplications. *Advanced Materials* **32**, 1802075 (2020).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization